

Omin Kwon

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RESEARCH INTEREST

Computer Architectures & Machine Learning Systems, with an emphasis on co-designing ML algorithms and computer systems for efficient ML.

EDUCATION

Seoul National University

Sept 2025 – present

M.S. in Computer Science and Engineering

- Advisor: Prof. Jae W. Lee, ARC Lab
- GPA: 4.3/4.3

Seoul National University

Mar 2019 – Aug 2025

B.S. in Electrical and Computer Engineering (summa cum laude)

- GPA: 4.26/4.3 (upper-division), 3.95/4.3 (overall)
- Leave of absence for mandatory military service (May 2021-Feb 2023)

PUBLICATIONS

HERALD: High Throughput Block Diffusion LLM Serving via CPU-GPU Cooperative KV Cache Retrieval

Omin Kwon, Doyeon Kim, Jongseok Park, Seung Yul Lee, Ion Stoica, Jae W. Lee

Under review

Paper Link: arxiv.org/abs/2606.21633

ELIXIR: Disaggregating FFN, Attention, and KV Cache via CXL-PNM for Elastic LLM Serving

Omin Kwon*, Kyungsoo Kim*, Ugyeong Song, Yeonhong Park, Nachuan Wang, Namsung Kim, Jae W. Lee

Under review

MAGE: All-[MASK] Block Already Knows Where to Look in Diffusion LLM

Omin Kwon, Yeonjae Kim, Doyeon Kim, Minseo Kim, Yeonhong Park, Jae W. Lee

ICML 2026 AdaptFM Workshop (Best Paper Award); extended version under review

Paper Link: arxiv.org/abs/2602.14209

STAR: Improving Lifetime and Performance of High-Capacity Modern SSDs Using State-Aware Randomizer

Omin Kwon*, Kyungjun Oh*, Jaeyong Lee, Myungsuk Kim, Jihong Kim

ICCAD 2025

Paper Link: doi.org/10.1109/ICCAD66269.2025.11240772

NestedFP: High-Performance, Memory-Efficient Dual-Precision Floating Point Support for LLMs

Haeun Lee, Omin Kwon, Yeonhong Park, Jae W. Lee

NeurIPS 2025

Paper Link: neurips.cc/virtual/2025/loc/san-diego/poster/117613

AiDE: Attention-FFN Disaggregated Execution for Cost-Effective LLM Decoding on CXL-PNM

KyungSoo Kim, Omin Kwon, Yeonhong Park, Jae W. Lee

IEEE Computer Architecture Letters (CAL)

Paper Link: doi.org/10.1109/LCA.2025.3597323

EXPERIENCES

Visiting Researcher

Mar 2026 – May 2026

University of Illinois Urbana-Champaign (UIUC)

Urbana, IL, USA

- FAST Lab, hosted by Prof. Nam Sung Kim

Microelectronics (430.207B) Tutor Seoul National University	Mar 2025 – June 2025 Seoul, South Korea
Digital System Design (430.315) Tutor Seoul National University	Sept 2024 – Dec 2024 Seoul, South Korea
Computer Organization and Design (430.322) Tutor Seoul National University	Mar 2024 – June 2024 Seoul, South Korea
Military Service Republic of Korea Air Force	May 2021 – Feb 2023 South Korea
PRESENTATIONS	
MAGE: All-[MASK] Block Already Knows Where to Look in Diffusion LLM Oral presentation at the ICML 2026 AdaptFM Workshop. arXiv.	2026
STAR: Improving Lifetime and Performance of High-Capacity Modern SSDs Using State-Aware Randomizer Oral presentation at ICCAD 2025, Munich. Paper.	2025
HONORS & AWARDS	
Best Paper Award @ ICML26 AdaptFM Workshop ICML 2026 AdaptFM Workshop 1 Best Paper among 123 accepted papers from 218 submissions.	July 2026
Grand Prize (Minister of Science and ICT Award), 2024 AI Semiconductor Idea and Design Contest Ministry of Science and ICT Designed a resolution-aware diffusion accelerator leveraging computational staleness. News: Edaily.	Jan 2025
Grand Prize (Deputy Prime Minister and Minister of Education Award), 2024 STOB League Semiconductor Contest Ministry of Education Designed a state-aware data randomizer for high-performance SSDs, improving performance and device lifetime. News: Dong-A Ilbo.	Dec 2024
Grand Prize, 2024 FriendlyAI LLM Hackathon FriendlyAI Developed a knowledge-graph RAG application using the FriendlyAI API.	July 2024
SNU Specialized Semiconductor College Scholarship Seoul National University	Aug 2024 – Aug 2025
Merit-Based Scholarship Seoul National University	2023, 2024